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Prospective associations of screen time at age 2 with specific behavioral subscales at age 3: a cohort study

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ABSTRACT

Background We aim to discover which, if any, of the subscales of internalizing and externalizing behavioral problems at age 3 are still associated with screen time (ST) at age 2 after adjusting for behavioral problems scores at age 2.

Methods This study was conducted under the Tohoku Medical Megabank Project Birth and Three-Generation Cohort Study. Information was gathered prospectively, with 7207 mother–child pairs included in the analysis. Children's ST was categorized in hours a day at age 2 (<1, 1-<2, 2-<4, ≥4). We assessed children's behavioral problems using the Child Behavior Checklist for Ages 1½–5 (CBCL) at ages 2 and 3. 'Having behavioral problems' was defined by them being within a clinical range for internalizing behaviors (withdrawn, somatic complaints, anxious/depressed and emotionally reactive) and externalizing behaviors (attention problems and aggressive behaviors) at age 3. Continuous scores on each of the behavioral problem scales at age 2 were used as covariates.

Results Greater ST for children at age 2 was associated with specific subscales for emotionally reactive and aggressive behaviors at age 3.

Conclusions This study found that ST is prospectively associated with some behavioral scales but not others.

Keywords childcare, developmental/behavioral health, epidemiology, screen time

Introduction

Screen time (ST) is the amount of time people spend watching screen devices, such as TV, video games, mobile phones, tablets, etc. The World Health Organization (WHO) and the American Academy of Pediatrics (AAP) have issued guidelines recommending ST limits for children[1,2]—1 h a day for children aged 2–5 years[2]—to ensure that children are getting enough physical activity and sleep for healthy growth. However, due to the rapid proliferation of digital devices and the COVID-19 pandemic, ST for children and adolescents has increased markedly[3–6] with a recent meta-analysis showing that only a minority of children meet the guidelines.[7] Given the above, examining how children's ST associates their health and development has become an urgent public health issue.

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Behavioral problems are one indicator of a child's psychosocial adjustment/maladjustment.[8] Children with chronic behavioral problems for internalizing behaviors, consisting of traits, such as 'withdrawn', 'somatic complaints', 'anxious/depressed' and 'emotionally reactive',[8,9] and externalizing behaviors, consisting of traits, such as 'attention problems' and 'aggressive behaviors'[8,9] have been reported to result in long-term economic and social disadvantage.[10] Previous studies have hypothesized that ST is associated with behavioral problems in children, and the associations examined[11–24]: cross-sectional studies,[11–16] prospective studies,[17–21] systematic reviews[22,23] and meta-analysis[24] have pointed to an association between ST and children's behavioral problems. However, three important issues need to be considered to more fully understand the association.

First, when examining the association between ST and behavioral problems, it is important to consider information on the child's behavior prior to or contemporaneous with the assessment of ST exposure in order to distinguish ST associations with later behavioral problems prospectively from long ST as a characteristic of children with behavioral problems. To our knowledge, only two studies have examined the association between ST and behavioral problems in this way.[19,20] Verlinden *et al.* examined the association of ST with the incidence of behavioral problems for externalizing behaviors. In this study, higher ST exposure at ages 2 and 3 was associated with a higher incidence of behavioral problems for externalizing behaviors at age 3.[19] Neville *et al.*[20] confirmed a bidirectional association between ST and internalizing behaviors at ages 3 and 5. Externalizing behaviors at age 3 were directionally associated with prolonged ST at age 5 by random-intercept cross-lagged panel models (RI-CLPM).[20] To further understand the association between ST and behavioral problems, studies updating the above two previous studies are warranted.

Second, we need to know which specific area of behavioral problems the ST is associated with. If an association between ST and internalizing and externalizing behaviors is confirmed, it would be important to clarify whether the association can be confirmed for all subscales or only a specific one, as each subscale has a different impact on mental health. Finding a subscale specifically associated with ST would clarify the association between ST and behavioral problems and potentially result in significant advances in treatment for parent, caregivers and pediatricians.

Given this circumstance, the purpose of this study was to examine the association of children's ST at age 2 with behavioral problems for internalizing and externalizing behaviors and subscales of internalizing and externalizing behaviors at age 3, considering each of the behavioral problem scores at age 2.

Methods

Study design and population

The present study was conducted under the Tohoku Medical Megabank Project Birth and Three-Generation Cohort Study (TMM BirThree Cohort Study). The details of the TMM BirThree Cohort Study are described elsewhere.[25–28] In brief, from July 2013 to March 2017, pregnant women from ∼50 obstetric clinics and hospitals in Miyagi and Iwate prefectures in Japan were recruited into the TMM BirThree Cohort Study.[25,26] Trained genome medical research coordinators explained the details of the study to all potential participants and obtained signed consent from all participants.[25,26] The protocol was reviewed and approved by the Institutional Review Board of the Tohoku Medical Megabank Organization (27 May 2013; Approval No: 2013-1-103-1). A total of 7207 mother–child pairs were included in the analysis (Fig. 1). A total of 23 730 mother–child/fetus pairs were excluded: withdrawn informed consent ($n = 615$), maternal multiple participation in the survey ($n = 911$), miscarriage or stillbirth ($n = 374$), multiple birth ($n = 539$), missing information on children's ST at age 2 ($n = 9421$), missing information on children's behavioral problems at age 2 ($n = 1707$), missing information on children's behavioral problems at age 3 ($n = 2942$) and children diagnosed with autism spectrum disorder (ASD), cerebral palsy or chromosomal abnormalities by age 3 ($n = 14$). The present study followed the Strengthening the Reporting of Observational Studies in Epidemiology (STROBE) reporting guideline.

Children's screen time

Children's ST at age 2 was assessed using a parent-completed questionnaire. It asked about children's ST as follows: 'On a typical day, how many hours do you allow your children to watch TV, DVDs, video games, internet games (including mobile phones and tablets), etc.?' The answer had five options: 'none', 'less than 1 hour (<1)', '1 hour or more but less than 2 hours ("1-<2")', '2 hours or more but less than 4 hours ("2-<4")' and '4 hours or more (≥4)'. Options 'none' and '<1' were merged, referring to the guidelines.[1,2] Four categories ('<1', '1-<2', '2-<4' and '≥4') were defined as exposure.

Children's behavioral problems

To assess children's behavioral problems at ages 2 and 3, we used the Child Behavior Checklist for Ages 1½–5 (CBCL),[8,9] which is a parent-completed screening tool consisting of

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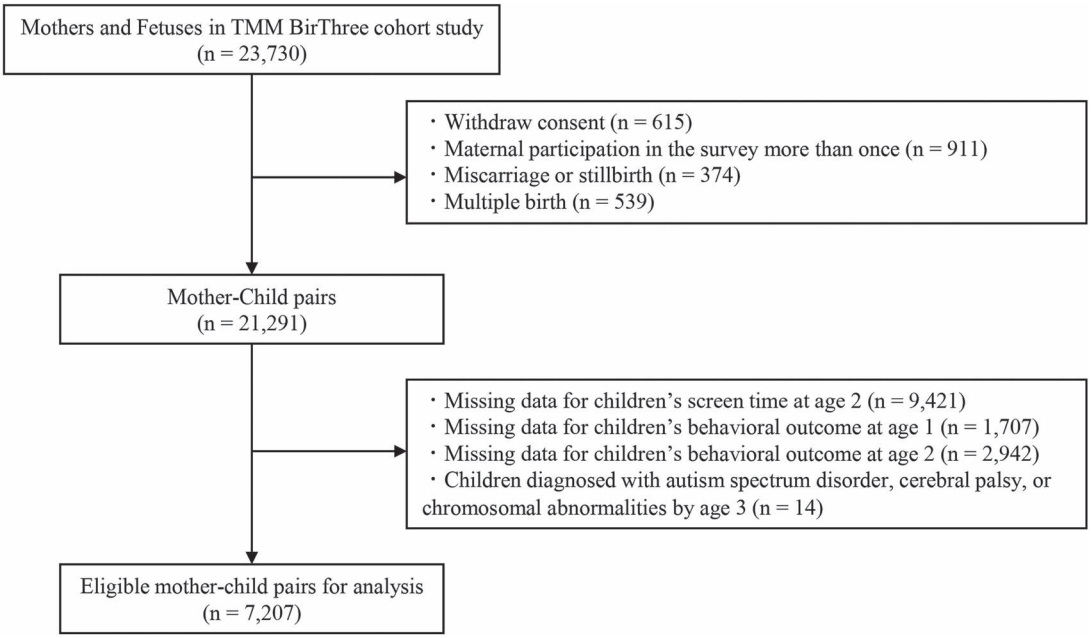

Fig. 1 Participant exclusion criteria in this study.

100 items. For each item, parents were asked to choose one answer from a three-point likert scale: 'does not apply' (0 points); 'somewhat applies' or 'sometimes applies' (1 point) and 'very applies' or 'often applies' (2 points). The 100 items of the CBCL were classified into seven subscales ('withdrawn', 'somatic complaints', 'anxious/depressed', 'emotionally reactive', 'attention problems', 'aggressive behaviors' and 'sleep problems').[8,9] The scale of internalizing behaviors was defined as the sum of answers to 'withdrawn', 'somatic complaints', 'anxious/depressed' and 'emotionally reactive'.[8,9] The scale of externalizing behaviors was defined as the sum of answers to 'attention problems' and 'aggressive behaviors'.[8,9] Since sleep problems are not a component of internalizing and externalizing behavior, this study was conducted on six subscales. In the previous study, [9] T-scores (mean: 50, standard deviation: 10) for each scale were calculated and standardized for Japanese children, with 64 or higher indicating a clinical range for externalizing and internalizing behaviors and 70 or higher indicating a clinical range for subscales.[9] In this study, the clinical range for externalizing and internalizing behaviors and their subscales were analyzed as outcomes. Continuous scores on the scale for each behavior at age 2 were used as covariates.

Covariates

Potential covariates that might influence the association between ST and behavioral problems in children were selected based on previous studies.[11–24,29–33] Information on the sex of the children was gathered from birth records. Information on maternal age at delivery was gathered from maternal medical records and divided into four categories (<25, 25–29, 30–34, ≥35 years). Information on household income (<4 000 000, 4 000 000–5 999 999 and ≥6 000 000 Japanese yen/year) was gathered from a questionnaire given the second trimester. Data on maternal educational attainment (high school graduate or less, junior college or vocational college graduate, university graduate or above) was gathered using a questionnaire given 1 year postpartum. Information on living with grandparents or other adults other than parents, living with younger siblings, living with older siblings, maternal psychological distress, maternal bonding disorder and each behavior assessed by the CBCL was gathered using a questionnaire given 2 years postpartum. Maternal psychological distress was assessed using the Japanese version of the K6,[34–36] with a total score of K6 ≥ 5 defined as psychological distress. The cut-off score was suggested to maximize the sum of specificity and sensitivity in the Japanese version of K6 in screening for mood and anxiety disorders.[36] Maternal bonding disorder was assessed using the Mother-to-Infant Bonding Scale Japanese version[37–39] (MIBS-J), with a total score of MIBS-J ≥ 5 defined as bonding disorder. A previous study of Japanese mothers with 1-month-old infants suggested that MIBS-J ≥ 5 had been correctly classified as ∼90% of pathological maternal bonding disorders.[39]

Statistical analysis

The characteristics of the participants were described according to four categories of ST at age 2. Characteristics were

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presented as frequencies with percentages and as medians with interquartile ranges (IQRs). In addition, because only ~33.8% ($n = 7207$) of the mother–child pairs who agreed to participate in the TMM BirThree cohort study ($n = 21\,291$) were included in our analysis, we performed a dropout analysis, a comparison of the characteristics of the participants in the analysis with those of the excluded participants, to examine external validity. The characteristics of the participants according to the two groups of participants included ($n = 7207$) and participants not included (n = 14 084) were described, and P values were obtained from chi-square tests comparing participant characteristics. The variables used in the dropout analysis were as follows: maternal age at delivery, maternal educational attainment, household income, maternal psychological distress at age 2, postpartum, maternal bonding disorder at age 2, postpartum, children's sex, living with grandparents or other adults other than parents at age 2, living with younger siblings at age 2 and living with older siblings at age 2. The association between ST at age 2 and the internalizing behaviors, externalizing behaviors and six subscales at age 3 were evaluated by multivariable logistic regression analysis to estimate odds ratios (ORs), 95% confidence intervals (CIs) and P value for trend (P for trend). 'ST < 1' was set as a reference. Covariates were adjusted for in three models: 'Model 1; Crude model', 'Model 2; Adjusted for maternal age at delivery, maternal educational attainment, household income, children's sex, living with grandparents or other adults other than parents at age 2, living with younger siblings at age 2, living with older siblings at age 2, maternal psychological distress at age 2, postpartum, and maternal bonding disorder at age 2, postpartum' and 'Model 3; Model 2 adjusted for continuous scores on each scale of behavioral problems at age 2 to consider the direction of the association'. Missing covariates were imputed through multiple imputation by chained equations using exposure, outcome, and covariates in each model.[40] Twenty sets of quasi-complete data were run through independent multivariable analyses, and the estimates were integrated.[40] All statistical analyses were performed using R, v.4.0.2, and 95% CIs not crossing 1.00 or P for trend < 0.05 were considered statistically significant.

Results

Characteristics of the study population

The participant characteristics according to the four categories of ST ('<1', '1-<2', '2-<4' and '≥4') at age 2 are shown in Table 1. Of the 7207 children assessed: 1497 (20.8%) to ST = '<1', 2739 (38.0%) to '1-<2', 2459 (34.1%) to '2-<4', and 512 (7.1%) to '≥4.' Children's behavioral problems at age 3 showed 616 (8.5%) had internalizing behaviors, and 880 (12.2%) had externalizing behaviors. The subscale of internalizing behaviors was assessed as 122 (1.7%) withdrawn, 174 (2.4%) somatic complaints, 132 (1.8%) anxious/depressed and 189 (2.6%) emotionally reactive. In the subscale of externalizing behaviors, 176 (2.4%) were screened in the clinical range for attention problems and 272 (3.8%) for aggressive behaviors. Children with higher ST were more likely to have characteristics, such as living with younger siblings, not living with grandparents or other adults other than parents, not living with older siblings, younger maternal age, lower household income, lower maternal education level, a higher prevalence of maternal psychological distress and a higher prevalence of maternal bonding disorder than the other groups.

Comparison of characteristics of included and not included participants

The results of the dropout analysis comparing the characteristics of included and not-included participants are shown in Supplementary Table 1. Mothers excluded from the analysis were younger, less educated, had lower incomes and had a higher prevalence of psychological distress and bonding disorder at 2 years postpartum. Children excluded from the analysis had a lower rate of living with older siblings.

Multivariable logistic regression analysis for children's ST and internalizing behaviors and the subscales

Table 2 shows the association between the four categories of children's ST ('<1', '1-<2', '2-<4' and '≥4') at age 2 with internalizing behaviors and these subscales at age 3 by multivariable logistic regression analysis using the 'ST < 1' category as reference. In model 2, an association was shown between children's ST at age 2 and internalizing behaviors (OR = 1.57; 95% CI: 1.20–2.06 for 2-<4 and 2.73; 1.95–3.82 for ≥4 versus < 1, P for trend < 0.001) a age 3. For the subscale of internalizing behaviors, associations were shown in withdrawn (2.60; 1.32–5.10 for ≥4 versus <1), somatic complaints (2.02; 1.12–3.64 for ≥4 versus <1), anxious/depressed (P for trend = 0.029) and emotionally reactive (2.67; 1.54–4.64 for ≥4 versus < 1, P for trend < 0.001). In model 3, after additional adjusting for scores on each scale at age 2 continuously, an association was shown between children's ST at age 2 and internalizing behaviors (1.98; 1.35–2.91 for ≥4 versus <1, P for trend = 0.001) and emotionally reactive (2.51; 1.39–4.54 for ≥4 versus <1, P for trend = 0.001).

Multivariable logistic regression analysis for children's ST and externalizing behaviors and the subscales

Table 3 shows the association between the 4 categories of children's ST at age 2 with externalizing behaviors and the subscales at age 3 by multivariable logistic regression analysis.

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Table 1 Characteristics of the participants according to the four categories of ST at age 2

	Total	*ST at age 26*			
	n = 7207	*<1*	*1-<2*	*2-<4*	*≥4*
		n = 1497	*n = 2739*	*n = 2459*	*n = 512*
Maternal age at delivery, *n* (%)					
<25 (y)	367 (5.1)	60 (4.0)	131 (4.8)	142 (5.8)	34 (6.6)
25–30 (y)	1749 (24.3)	319 (21.3)	636 (23.2)	636 (25.9)	158 (30.9)
30–35 (y)	2761 (38.3)	585 (39.1)	1066 (38.9)	926 (37.7)	184 (35.9)
>35 (y)	2330 (32.3)	533 (35.6)	906 (33.1)	755 (30.7)	136 (26.6)
Household income[a], *n* (%)					
<4 000 000 (JPY/y)	2290 (31.8)	395 (26.4)	805 (29.4)	890 (36.2)	200 (39.1)
4 000 000–6 000 000 (JPY/y)	2339 (32.5)	462 (30.9)	902 (32.9)	785 (31.9)	190 (37.1)
>6 000 000 (JPY/y)	2239 (31.1)	572 (38.2)	915 (33.4)	657 (26.7)	95 (18.6)
Missing	339 (4.7)	68 (4.5)	117 (4.3)	127 (5.2)	27 (5.3)
Maternal educational attainment, *n* (%)					
High school graduate or less	2024 (28.1)	329 (22.0)	722 (26.4)	780 (31.7)	193 (37.7)
Junior college or vocational college graduate	2590 (35.9)	524 (35.0)	1015 (37.1)	873 (35.5)	178 (34.8)
University graduate or above	2042 (28.3)	524 (35.0)	806 (29.4)	614 (25.0)	98 (19.1)
Missing	551 (7.6)	120 (8.0)	196 (7.2)	192 (7.8)	43 (8.4)
Maternal psychological distress at age 2 postpartum, *n* (%)					
Yes	1944 (27.0)	355 (23.7)	680 (24.8)	725 (29.5)	184 (35.9)
Missing	24 (0.3)	6 (0.4)	11 (0.4)	5 (0.2)	2 (0.4)
Maternal bonding disorder at age 2, postpartum, *n* (%)					
Yes	1413 (19.6)	253 (16.9)	493 (18.0)	525 (21.4)	142 (27.7)
Missing	93 (1.3)	21 (1.4)	31 (1.1)	33 (1.3)	8 (1.6)
Children's sex, *n* (%)					
Male	3715 (51.5)	782 (52.2)	1428 (52.1)	1241 (50.5)	264 (51.6)
Living with grandparents or other adults other than parents at age 2, *n* (%)					
Yes	1577 (21.9)	346 (23.1)	608 (22.2)	517 (21.0)	106 (20.7)
Living with younger siblings at age 2, *n* (%)					
Yes	709 (9.8)	94 (6.3)	249 (9.1)	283 (11.5)	83 (16.2)
Living with older siblings at age 2, *n* (%)					
Yes	3650 (50.6)	899 (60.1)	1421 (51.9)	1131 (46.0)	199 (38.9)
Behavioral problem scores at age 2, median (IQR)					
Internalizing behaviors	2.0 (0.0–4.0)	1.0 (0.0–3.0)	2.0 (0.0–4.0)	2.0 (1.0–5.0)	3.0 (1.0–6.0)
Withdrawn	0.0 (0.0–1.0)	0.0 (0.0–0.0)	0.0 (0.0–1.0)	0.0 (0.0–1.0)	0.0 (0.0–1.0)
Somatic complaints	0.0 (0.0–1.0)	0.0 (0.0–1.0)	0.0 (0.0–1.0)	0.0 (0.0–1.0)	1.0 (0.0–2.0)
Anxious/depressed	1.0 (0.0–2.0)	0.0 (0.0–1.0)	1.0 (0.0–2.0)	1.0 (0.0–2.0)	1.0 (0.0–2.0)
Emotionally reactive	0.0 (0.0–1.0)	0.0 (0.0–1.0)	0.0 (0.0–1.0)	0.0 (0.0–1.0)	0.0 (0.0–1.0)
Externalizing behaviors	5.0 (1.0–10.0)	3.0 (1.0–8.0)	5.0 (2.0–9.0)	6.0 (2.0–11.0)	8.0 (4.0–14.0)
Attention problems	1.0 (0.0–2.0)	0.0 (0.0–1.0)	1.0 (0.0–2.0)	1.0 (0.0–2.0)	2.0 (0.0–3.0)
Aggressive behaviors	4.0 (1.0–8.0)	3.0 (1.0–6.0)	4.0 (1.0–7.0)	5.0 (2.0–9.0)	6.0 (3.0–11.0)
Behavioral problems at age 3, *n* (%)					
Internalizing behaviors	616 (8.5)	79 (5.3)	196 (7.2)	248 (10.1)	93 (18.2)
Withdrawn	122 (1.7)	17 (1.1)	43 (1.6)	41 (1.7)	21 (4.1)
Somatic complaints	174 (2.4)	25 (1.7)	62 (2.3)	63 (2.6)	24 (4.7)
Anxious/depressed	132 (1.8)	19 (1.3)	40 (1.5)	57 (2.3)	16 (3.1)
Emotionally reactive	189 (2.6)	25 (1.7)	51 (1.9)	79 (3.2)	34 (6.6)
Externalizing behaviors	880 (12.2)	112 (7.5)	288 (10.5)	373 (15.2)	107 (20.9)
Attention problems	176 (2.4)	14 (0.9)	59 (2.2)	78 (3.2)	25 (4.9)
Aggressive behaviors	272 (3.8)	27 (1.8)	87 (3.2)	113 (4.6)	45 (8.8)

[a]To convert Japanese yen to US dollars, multiply by 0.0071.

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Table 2 Association of screen time at age 2 with internalizing behaviors and its subscales at age 3

Behavioral problems	*Screen time at age 2 (h)*	*Model 1*	*Model 2*	*Model 3*
		ORs (95%CI)	*ORs (95%CI)*	*ORs (95%CI)*
Internalizing behaviors	<1	Ref	Ref	Ref
	1-<2	1.38 (1.06–1.82)	1.21 (0.92–1.60)	1.16 (0.85–1.57)
	2-<4	2.01 (1.56–2.63)	1.57 (1.20–2.06)	1.28 (0.95–1.73)
	≥4	3.98 (2.90–5.49)	2.73 (1.95–3.82)	1.98 (1.35–2.91)
	P for trend	<0.001	<0.001	0.001
Withdrawn	<1	Ref	Ref	Ref
	1-<2	1.39 (0.80–2.51)	1.28 (0.72–2.27)	1.47 (0.79–2.75)
	2-<4	1.48 (0.85–2.68)	1.20 (0.67–2.15)	1.11 (0.58–2.10)
	≥4	3.72 (1.95–7.20)	2.60 (1.32–5.10)	1.85 (0.86–3.95)
	P for trend	0.002	0.054	0.512
Somatic complaints	<1	Ref	Ref	Ref
	1-<2	1.36 (0.86–2.22)	1.20 (0.75–1.93)	1.28 (0.78–2.10)
	2-<4	1.55 (0.98–2.52)	1.22 (0.76–1.97)	1.13 (0.68–1.87)
	≥4	2.90 (1.63–5.13)	2.02 (1.12–3.64)	1.72 (0.92–3.19)
	P for trend	0.001	0.062	0.299
Anxious/ depressed	<1	Ref	Ref	Ref
	1-<2	1.15 (0.67–2.04)	1.02 (0.58–1.78)	1.00 (0.55–1.82)
	2-<4	1.85 (1.12–3.20)	1.46 (0.85–2.49)	1.20 (0.67–2.15)
	≥4	2.51 (1.26–4.92)	1.79 (0.89–3.59)	1.38 (0.65–2.93)
	P for trend	<0.001	0.029	0.299
Emotionally reactive	<1	Ref	Ref	Ref
	1-<2	1.12 (0.70–1.84)	0.95 (0.58–1.55)	0.96 (0.57–1.62)
	2- <4	1.95 (1.26–3.14)	1.46 (0.91–2.32)	1.28 (0.77–2.12)
	≥4	4.19 (2.48–7.16)	2.67 (1.54–4.64)	2.51 (1.39–4.54)
	P for trend	<0.001	<0.001	0.001

Model 1: Crude model.

Model 2: Adjusted for maternal age at delivery, maternal educational attainment, household income, maternal psychological distress at age 2, postpartum, maternal bonding disorder at age 2, postpartum, children's sex, living with grandparents or other adults other than parents at age 2, living with younger siblings at age 2, and living with older siblings at age 2.

Model 3: Additionally adjusted for continuous scores on each scale of behavioral problems at age 2.

In Model 2, an association was shown between children's ST at age 2 and externalizing behaviors (OR = 1.35; 95% CI: 1.07–1.71 for 1-<2, 1.86; 1.48–2.34 for 2-<4 and 2.37; 1.75–3.21 for ≥4 versus <1, *P* for trend < 0.001). For the subscales of externalizing behaviors, associations were shown in attention problems (1.99; 1.10–3.61 for 1-<2, 2.46; 1.37–4.41 for 2-<4 and 3.04; 1.53–6.03 for ≥4 versus <1, *P* for trend < 0.001) and aggressive behaviors (1.64; 1.05–2.55 for 1-<2, 2.09; 1.35–3.23 for 2-<4 and 3.48; 2.09–5.80 for ≥4 versus <1, *P* for trend < 0.001). In model 3, after additional adjusting for scores on each of the scales at age 2 continuously, an association was shown between children's ST at age 2 and externalizing behaviors (*P* for trend = 0.033), and the subscale for aggressive behaviors (2.35; 1.34–4.15 for ≥4 versus <1, *P* for trend = 0.012).

Discussion

Main finding of this study

The present study shows that ST at age 2 was associated with internalizing and externalizing behaviors and the subscales for emotionally reactive and aggressive behaviors at age 3, even after adjusting for each of the behavioral problem scores at age 2.

What is already known on this topic and what this study adds

These results indicate that the directional association between ST and internalizing behaviors identified in the previous study by Neville *et al.* might have been skewed by the association between ST and emotional reactive,[20] and the association

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Table 3 Association of screen time at age 2 with externalizing behaviors and its subscales at age 3

Behavioral problems	*Screen time at age 2 (h)*	*Model 1*	*Model 2*	*Model 3*
		ORs (95%CI)	*ORs (95%CI)*	*ORs (95%CI)*
Externalizing behaviors	<1	Ref	Ref	Ref
	1-<2	1.45 (1.16–1.83)	1.35 (1.07–1.71)	1.16 (0.89–1.52)
	2-<4	2.21 (1.78–2.77)	1.86 (1.48–2.34)	1.29 (0.99–1.68)
	≥4	3.27 (2.45–4.35)	2.37 (1.75–3.21)	1.32 (0.93–1.89)
	P for trend	<0.001	<0.001	0.033
Attention problems	<1	Ref	Ref	Ref
	1-<2	2.33 (1.34–4.36)	1.99 (1.10–3.61)	1.77 (0.93–3.37)
	2-<4	3.47 (2.02–6.41)	2.46 (1.37–4.41)	1.88 (0.998–3.53)
	≥4	5.44 (2.84–10.82)	3.04 (1.53–6.03)	1.76 (0.83–3.71)
	P for trend	<0.001	<0.001	0.155
Aggressive behaviors	<1	Ref	Ref	Ref
	1-<2	1.79 (1.17–2.81)	1.64 (1.05–2.55)	1.54 (0.94–2.51)
	2-<4	2.62 (1.74–4.09)	2.09 (1.35–3.23)	1.53 (0.95–2.47)
	≥4	5.25 (3.24–8.65)	3.48 (2.09–5.80)	2.35 (1.34–4.15)
	P for trend	<0.001	<0.001	0.012

Model 1: Crude model.

Model 2: Adjusted for maternal age at delivery, maternal educational attainment, household income, children's sex, living with grandparents or other adults other than parents at age 2, living with younger siblings at age 2, living with older siblings at age 2, maternal psychological distress at age 2, postpartum, and maternal bonding disorder at age 2, postpartum.

Model 3: Additionally adjusted for continuous scores on each scale of behavioral problems at age 2.

between ST and incidence of behavioral problems for externalizing behaviors identified in the study by Verlinden *et al.*[19] skewed by the association between ST and aggressive behaviors. We hypothesize that long ST is an identifiable prospectively associated factor for emotionally reactive or aggressive behaviors. A previous study showed that frequent use of screen devices to calm children aged 3–5 years was associated with increased emotional dysregulation in children and discussed whether the use of screen devices might deprive children of the opportunity to learn emotional control skills.[41] Regarding the directional association of ST and aggressive behaviors, previous studies have discussed the hypothesis that exposure to violent content by screen devices could lead to aggressive behaviors in children.[42–45] If ST is the cause of behavioral problems for emotional reactive and aggressive behaviors, then refraining from giving ST to children who tend to have aggression and emotional reactive behavior may prospectively limit the severity of subsequent behavioral problems.

To consider whether long ST is a characteristic of children with behavioral problems or whether ST is longitudinally associated with subsequent behavioral problems, Model 3 adjusted for each scale score of behavioral problems at age 2 in addition to the adjustments in Model 2. With these additional adjustments, the associations on the subscale of withdrawn, somatic complaints, anxious/depressed, and attention problems that had been identified in the Model 2 were no longer identified. A possible explanation for this could be that long ST might be characteristic of children who had behavioral problems in the four subscales for which the association disappeared. In support of this, studies have found that the genetic risk for ASD and attention-deficit/hyperactivity disorder (ADHD) is associated with long ST, suggesting that long ST may be a possible characteristic of ASD or ADHD.[46,47] Thus, it cannot be ruled out that children in the present study with congenital behavioral problems—withdrawn, somatic complaints, anxious/depressed and attention problems—may have had long ST characteristics. It is important to interpret the results of the present study not only considering that ST is longitudinally associated with specific subscales of internalizing or externalizing behaviors but also that it may be a characteristic of children with behavioral problems on specific subscales.

In this study, 20.8% of the children met the ST guidelines recommendation[1,2] (ST < 1) at age 2, while 79.2% did not. In this era of rapid proliferation of digital devices, it is not easy to limit the use of digital devices to 1 h. Further, there is a tendency for ST to focus only on the negative aspects

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of children's development. Indeed, the UK report states that 'the ST advice to parents remains overwhelmingly focused on risk and harm, with little focus on opportunities to "learn, connect and create"'.[48] However, there are good aspects of ST for children's development. A meta-analysis suggests that the longer the non-educational ST is associated with lower language skills, and the longer the ST with an educational program is associated with the higher language skills.[49] It is realistic to identify and limit aspects of ST that negatively affect children's development while utilizing educational aspects in today's digital era.

Limitations and strength of this study

This study has some limitations. First, the information collected did not allow us to distinguish between educational STs and non-educational STs. In future studies, a detailed examination of the associations of both screen content (e.g. educational versus non-educational and violent versus non-violent) and ST with subscales of behavioral problems will contribute to more detailed mechanistic considerations. Second, mother–child pairs who agreed to participate in the TMM BirThree cohort study, only ∼33.8% were included in our analysis, which affects the external validity of our results. In fact, the dropout analysis confirmed differences in the characteristics of included and not included participants (Supplementary Table 1). The characteristics of the excluded mother–child pairs were similar to those of the children with longer ST and a higher prevalence of behavior problems (Table 1 and Supplementary Table 1). Under these circumstances, the results of this study may be an underestimate of the association between ST and behavioral problems in children. Future studies analyzing populations with a variety of characteristics will provide a more detailed picture of the overall association between ST and behavioral problems. Third, the study relied on parent-reported estimates of their children's ST. Although this may be subject to recall and social desirability bias, self-report is currently the better method for assessing ST for a large population,[6,7] given its simplicity and invasiveness. In the future, more objective measures, such as device monitoring, may provide more accurate ST data. Fourth, this study's results are for a Japanese population at that time and cannot be fully adapted to other foreign cultures and environments, including the radical proliferation of screen devices and popular content on screen devices. Fifth, we have done our best to adjust for confounding factors. However, the nature of observational studies makes it difficult to completely adjust unmeasured confounding factors. Finally, because we were unable to examine the content of ST, such as educational or non-educational content, we avoided interpreting the odds ratios in a public health framework. Further considerations in a cohort study collecting the content of ST information are awaited. On the other hand, one of the strengths of this study is that the analyses were conducted in a prospective cohort study with a collection of various variables longitudinally. Moreover, by focusing on subscales of internalizing and externalizing behaviors, we have suggested for the first time that there are subscales specifically and prospectively associated with ST.

Conclusion

In conclusion, greater ST for children at age 2 was associated with specific subscales of behavioral problems for emotionally reactive and aggressive behaviors at age 3. These associations remained even after adjusting for each of the behavioral problem scores at the time of the ST measurement.

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Supplementary data

Supplementary data are available at the *Journal of Public Health* online.

Conflict of interest

All authors have no competing interests to declare.

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Ethics approval and consent to participate

The TMM BirThree Cohort Study protocol was reviewed and approved by the Ethics Committee of Tohoku University Tohoku Medical Megabank Organization (27 May 2013; Approval No: 2013-1-103-1). The methods in this study have been carried out in accordance with the Declaration of Helsinki. All participants received explanations from trained genome medical research coordinators and provided written informed consent.

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Consent for publication

Not applicable.

Data availability

The data that support the findings of this study are available from the TMM biobank; however, restrictions apply to the availability of these data, which were used under license for the current study and hence are not publicly available. Data are available from the authors upon reasonable request and with the permission of the TMM biobank. All inquiries about access to the data should be sent to the TMM biobank (dist@megabank.tohoku.ac.jp).

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